Argus Master CAM CNC Software

Client Functional Verification & Interactive Step-by-Step Testing Guide

Software Release: v2.4.0 Production Build	**Application Target:** DMG MORI DMU 50 (5-Axis) / VMC 3-Axis
Execution Mode: 100% Dynamic Math (Zero Mock Flow)	**Primary Default Theme:** Mastercam Light CAD Palette
Tested Environment: Windows 10/11 x64 Desktop	**Core Stack:** React 19, Three.js WebGL, Rust Native (Tauri v2)

1. Executive Testing Scope & Dynamic Architecture

This verification document guides client evaluation engineers and CAM programmers through the operational capabilities of **Argus Master CAM**. Every capability demonstrated herein operates on **real geometric mathematics, live 3D kinematic trajectories, and physical cutting force algorithms**. No mock paths or pre-recorded flows are utilized.

GUARANTEED DYNAMIC CAPABILITIES READY FOR YOUR EVALUATION:

- **Mathematical Toolpath Generation:** Dynamic Pocketing, Facing, Waterline, Swarf 5-Axis, Rest Milling, and Pencil Tracing compute real cutter contact points from stepover % and stepdown depths.
- **Interactive 3D Backplot:** Three.js kinematic cutter assembly navigates real 3D trajectories with live X/Y/Z coordinate telemetry and solid material removal.
- **Multi-Dialect G-Code Post-Processor:** Generates and exports syntactically compliant CNC programs for Siemens 840D, Heidenhain TNC 640, Fanuc 31i, and Haas NGC.
- **Machining Physics Engine:** Calculates Kienzle cutting force (Fc), spindle power (kW), torque (Nm), and cantilever deflection (µm) using ISO material standards.

2. How to Launch the Software on Your Test Station

Open PowerShell or Command Prompt in the application folder and choose your preferred execution method:

Native Desktop Window (Recommended): `npm run tauri dev` *Launches the hardware-accelerated Windows native desktop binary.*	**Browser Mode (Alternative):** `npm run dev` *Spins up the Vite engine at http://localhost:5173 for quick web inspection.*

3. Overview of Test Scenarios in this Guide

This guide is divided into five targeted functional test modules with step-by-step instructions and expected outputs:

Scenario 1: 3D Toolpath Backplot & Kinematic Verification	Interactive tool tracing, feed/rapid color tracking & solid verify
Scenario 2: Dynamic CAM Operations Wizard	Real-time toolpath recalculation via stepover % and stepdown changes
Scenario 3: 5-Axis G-Code Post & File Download	DMU 50 stroke audit & local download of .nc / .mpf / .h programs
Scenario 4: Feeds, Speeds & Kienzle Physics	Cutting force, spindle load kW, torque & tool deflection live math
Scenario 5: 3D Collision Audit & CPlanes Studio	Holder interference check, 5-axis angled planes & light/dark toggle

TEST SCENARIO 1: In-Viewport 3D Backplot & Kinematic Verification

Target Capability: Validate that toolpaths are true 3D spline trajectories and that the kinematic cutter assembly moves accurately with live telemetry.

#	Action to Perform	Expected System Response & Validation Criteria
1.1	Inspect the left **Operations Manager** sidebar.	A structured tree displays the active CAM operations: *Face Milling (Top Z:0)*, *2D Dynamic Adaptive Pocket*, *3D Waterline Finishing*, and *5-Axis Swarf*. Each card displays tool number, badge, and visibility state.
1.2	In the top Ribbon bar, open the **Toolpaths** tab and click **Backplot** (cyan icon) — OR click the **Play icon** on Operation #2 in the sidebar.	The 3D Viewport switches into Backplot mode. A 3D cutter assembly (fluted endmill + cylindrical shank + ISO-standard tapered toolholder) snaps to the trajectory start point.
1.3	Click the **Play (■)** button on the floating HUD control bar at the bottom of the viewport.	The cutter assembly navigates along the toolpath in continuous motion. Cutting feed passes (G01/G02/G03) render in emerald/blue; rapid positioning moves (G00) render in dashed amber lines.
1.4	Drag the **Timeline Scrubber Slider** back and forth manually.	The tool assembly tracks immediately to the exact percentage position. The HUD telemetry panel displays real-time **X, Y, Z coordinates (mm)** and cutting feed status.
1.5	Switch the mode toggle on the scrubber from **Backplot** to **Verify Solid**.	Solid workpiece material removal activates, dynamically carving the raw stock billet to reveal machined pocket cavities and contours.

TEST SCENARIO 2: Dynamic CAM Toolpath Generation & Editing

Target Capability: Prove that toolpaths recalculate mathematically on-the-fly when user machining parameters change.

#	Action to Perform	Expected System Response & Validation Criteria
2.1	Click the green **Operations Wizard** button in the Toolpaths ribbon.	The Mastercam CAM Operations Manager Wizard modal opens displaying strategy options (Facing, Pocket, Waterline, Swarf, Adaptive, Rest, Pencil).
2.2	In the **Create Operation** tab, select **Dynamic Pocket**.	Form fields populate with parameter inputs: Assigned Tool, Stepover %, Stepdown, Total Depth, Stock to Leave (XY & Z), and Rapid Clearance Height.
2.3	Modify the cutting geometry: • Change **Stepdown** from 4.0 mm to 1.5 mm • Change **Stepover %** from 50% to 25%	Input fields update instantly with real-time numeric validation.
2.4	Click **Generate & Add Operation**.	The modal closes. **Inspect the 3D Viewport:** The toolpath instantly regenerates with noticeably denser, finer z-level depth passes matching the exact 1.5mm stepdown and 25% stepover.
2.5	Click the **Eye icon** on any operation in the left sidebar.	The selected toolpath pass toggles between visible and hidden immediately in the 3D scene.

TEST SCENARIO 3: 5-Axis G-Code Post-Processor & Real File Download

Target Capability: Verify real multi-controller NC program generation, kinematic stroke auditing, and local file export.

#	Action to Perform	Expected System Response & Validation Criteria
3.1	In the Ribbon (Toolpaths tab), click the highlighted **G1 Post** button (or **Post & Export** in Machine tab).	The **NC2Check 5-Axis Post-Processor & Kinematic Audit** studio modal opens.
3.2	Select different **CNC Controller Dialects** from the dropdown: • **Siemens 840D sl / Sinumerik ONE** (`.mpf`) • **Heidenhain TNC 640 (PLANE SPATIAL)** (`.h`) • **Fanuc 31i-B5** (`.nc`) • **Haas NextGen (NGC)** (`.nc`)	The post engine dynamically reconfigures program headers, block numbering, tool change syntax (`M06`), plane spatial rotations, and canned cycle parameters.
3.3	Click **Run Pre-Flight Audit & Generate G-Code**.	The audit engine tests toolpaths against DMG MORI DMU 50 physical travel limits (X: ±250mm, Y: ±200mm, Z: -100/+350mm, B: -5°/110°). Success card displays: **Line Count, Total Cutting Distance (m), and Estimated Machining Time (min)**.
3.4	Click **Download File**.	A real CNC program file (e.g. DMU50_ZeroClamp_Job.nc) is downloaded to your workstation. Open it in any text editor — it contains complete, syntactically valid G-code ready for machine transfer.

TEST SCENARIO 4: Feeds, Speeds & Kienzle Cutting Physics Studio

Target Capability: Validate that scientific cutting forces, spindle power, torque, and tool deflection are computed live.

#	Action to Perform	Expected System Response & Validation Criteria
4.1	In the Ribbon (Machine or Toolpaths tab), click the cyan **Feeds & Speeds** button.	The Feeds, Speeds & Machining Physics Calculator opens.
4.2	Change the **Workpiece Material** dropdown from *Aluminum 6061-T6* to *Titanium Ti-6Al-4V* or *Inconel 718*.	Physical telemetry cards update live: **Kienzle Specific Force (kc1.1)**, **Spindle Power (kW)**, and **Torque (Nm)** increase in direct accordance with material hardness.
4.3	Modify **Tool Stickout Length** (e.g. increase from 45mm to 70mm).	The **Tool Cantilever Deflection (μm)** recalculates using cubic cantilever beam deflection equations (y = F*L^3 / 3*E*I). Automatic high-deflection alerts trigger if deflection exceeds tolerance.
4.4	Click **Apply to Active Tool** or **Apply to Operation**.	The calculated spindle RPM, feed rate, and depths are pushed directly into the tool library and active operation parameters.

TEST SCENARIO 5: 3D Collision Audit, CPlanes & Theme Engine

Target Capability: Test kinematic holder interference verification, dynamic 5-axis coordinate planes, and 1-click theme switching.

#	Action to Perform	Expected System Response & Validation Criteria
5.1	In the Ribbon (Toolpaths tab), click **Collision Check** (under Safety).	The 3D Gouge & Kinematic Interference engine audits cutter flutes, shank, and holder against the workpiece mesh. Collision points are highlighted with a 1-click **Auto-Fix Clearances** button.
5.2	In the Ribbon (Home or Machine tab), click **CPlane Studio**.	The CPlanes & Dynamic WCS Origin Studio opens.
5.3	Under **Angled Plane**, enter Tilt B = 30°, Rotation C = 45°, and click **Create & Activate Plane**.	**Inspect the 3D Viewport:** A dynamic construction plane grid with an RGB orientation triad renders tilted at the exact 30°/45° compound angle.
5.4	Click the **Sun / Moon icon** in the top TitleBar (quick access area).	The entire interface flips instantly between **Mastercam Light CAD** and **Industrial Dark CAD** with zero latency or state loss.

4. Client Test Verification Checklist & Sign-Off Sheet

Please record evaluation results during your testing session and return this page to Argus Technologies:

Feature / Module Under Test	Evaluation Criteria	Status	Client Notes / Comments
1. In-Viewport 3D Backplot	Smooth tool animation with coordinate telemetry & scrubber control	[ ] PASS [ ] FAIL	
2. Volumetric Solid Verify	Workpiece stock cavity removes accurately along toolpath pass	[ ] PASS [ ] FAIL	
3. Dynamic Toolpath Recalc	Changing stepdown/stepover regenerates 3D passes dynamically	[ ] PASS [ ] FAIL	
4. Multi-Dialect NC Post	Siemens / Fanuc / Heidenhain G-code generated and downloaded	[ ] PASS [ ] FAIL	
5. Feeds & Speeds Physics	Kienzle Fc, power kW, torque, and deflection calculate live	[ ] PASS [ ] FAIL	
6. 5-Axis CPlanes & Theme	Tilted 3D grid renders; 1-click Light/Dark toggle transitions cleanly	[ ] PASS [ ] FAIL	

Client Evaluator Name: _____	**Job Title / Role:** _____
Company / Organization: _____	**Evaluation Date:** _____
Overall Evaluation: [ ] ACCEPTED [ ] CONDITIONAL [ ] REVISE	**Signature:** _____

Technical Inquiries & Feedback Submission: Contact the engineering team at **engineering@argus-technologies.internal**. All CNC post-processors and kinematic digital twin models can be tailored to your shop's specific CNC machine configurations.